CODE SPORT

Sandya Mannarswamy

In this month's column, we feature a set of questions on algorithms, data structures and operating systems.

Given that we have a number of requests from our student readers who are preparing for on-campus interviews, let's continue our discussion on interview questions from last month's column. The focus is on questions regarding algorithms, data structures and operating systems.

1. We know that searching a non-sorted array of N integer elements has a time complexity of *O(N).* What is the time complexity in searching for an element in the array if the array is completely sorted? Given that the array is mostly sorted with a few elements being out of place, what would be the time complexity to search for an element in the array? Can you do this in *O(logN)*? If so, how? If not, give the reason why it is not possible.
2. The lowest common ancestor (LCA) of two nodes 'u' and 'v' in a binary search tree is defined as node A which has both 'v' and 'w' as its descendants (note that a node can be its own descendant). Write an algorithm to find the lowest common ancestor of a binary search tree.
3. Consider the question (2) above. Instead of writing a new algorithm for finding the LCA, if you are given the in-order traversal of the binary search tree along with the nodes 'u' and 'v' for which the LCA needs to be found, can you use the in-order traversal information to determine the LCA? Does this work in all cases? If so, explain why? If not, give a counter-example.
4. Can you identify what is the bug in the following code snippet:

```
#include <alloca.h>

char* foobar(int sz) {
char* ptr = alloca(sz);
return ptr;
}
int main() {
char* ptr = foobar(20);
 *ptr = 'a';
 printf("%c\n", *ptr);
}
```

5. You are given a set 'A' containing N integer elements. You are asked to solve the problem of finding the minimum in a given range of A. For instance, if A is {8, 2, 9, 5,1,4, 11, 3, 14}, range minimum for A[1…5] is given by the A[5] since A[5] is 1 and that is the minimum element in the range A[1..5]. Can you come up with an O(N) algorithm for solving the range minimum query if you are allowed to preprocess the set A?
6. Can you explain how a system call made by an application code actually enters the kernel code? What is the X86 architectural support that is provided, which is made use of by the system call entry point in Linux?
7. You are asked to add a brand new system call to the Linux kernel. Explain what parts of the kernel code you would need to modify and why?
8. Can you identify the problem with the following recursive code:

```
int find_factorial(int number) {
    return number * find_factorial(number - 1);
}
```

   Is it always possible to rewrite a recursive code to non-recursive code? If not, can you give a counter-example?
9. We are all familiar with merge-sort. Given an array of N integer elements, we break up the array into two halves, sort each half separately and then merge the two sorted halves. This is a classic example of the 'Divide and Conquer' approach, which divides the problem into several sub-problems, solves each of the sub-problems